

Kinetics of Fluorapatite Precipitation in Fluoride-Contaminated Water

Published as part of ACS ES&T Engineering special issue "Celebrating the 25th Anniversary of the Chemical Research Society of India".

Aravinth Siva Subramaniam Ekamparam, Surya Sujathan, and Abhas Singh*



Cite This: <https://doi.org/10.1021/acsestengg.4c00235>



Read Online

ACCESS |

Metrics & More

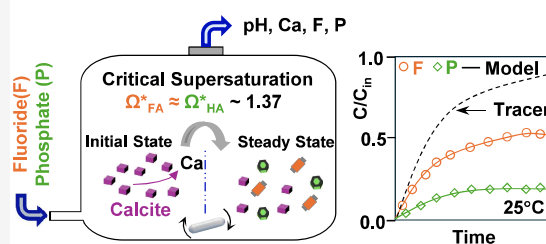
Article Recommendations

Supporting Information

ABSTRACT: Calcium phosphate (Ca-PO_4) solids, especially apatites, are considered potential amendments to remediate aquifers impacted by inorganic pollutants such as fluoride (F) due to their low solubilities. However, the precipitation kinetics of these solids at room temperature are poorly constrained, which limits their application. Specifically, to evaluate the extent and kinetics of F remediation through fluorapatite (FA; $\text{Ca}_5(\text{PO}_4)_3\text{F}$) precipitation, the quantification of the fluorapatite formation constant is important. In this contribution, flow-through experiments were conducted to estimate the surface-normalized rate constant of FA precipitation. Experiments were performed in the presence of a typical Ca source, calcite ($4 \text{ g}\cdot\text{L}^{-1}$), with a fixed influent phosphate dosage (1 mM) at variable influent F concentrations ($0.1\text{--}0.42 \text{ mM}$) and flow rates ($0.01\text{--}0.04 \text{ mL}\cdot\text{min}^{-1}$). Reactor effluents were collected using a fraction collector at specific time intervals and analyzed for pH and concentrations of F, Ca, and PO_4 until a steady-state was reached. The reacted solids collected after the experiments were characterized for the identification of precipitated phases. Process-based elemental mass balance equations were used to model the eluent concentrations. The surface normalized rate constants of precipitation of fluorapatite and hydroxyapatite (HA; $\text{Ca}_5(\text{PO}_4)_3\text{OH}$) were determined to be $10^{-28.24 \pm 1.24} \text{ }\mu\text{moles}\cdot\text{m}^{-2}\cdot\text{min}^{-1}$ and $10^{-21.74 \pm 6.85} \text{ }\mu\text{moles}\cdot\text{m}^{-2}\cdot\text{min}^{-1}$, respectively. However, the modeled critical supersaturation ratio for fluorapatite was comparable to the ratio for hydroxyapatite. Quantification of these fundamental rate constants fill a key knowledge gap that would likely help in predicting the fate and transport of F and PO_4 in contaminated aquifers or in the treatment of F-contaminated surface and aquifer waters.

KEYWORDS: fluoride, fluorapatite, hydroxyapatite, precipitation, kinetics

Fluorapatite (FA; ) and Hydroxyapatite (HA; ) Precipitation rate constants ($\mu\text{moles}\cdot\text{m}^{-2}\cdot\text{min}^{-1}$): $k_{\text{FA}} \sim 10^{-28.2}$; $k_{\text{HA}} \sim 10^{-21.7}$



INTRODUCTION

High fluoride (F) concentrations above the permissible drinking water limit ($1.5 \text{ mg}\cdot\text{L}^{-1}$)¹ have been reported in the groundwater of several parts of the world.^{2–6} Currently, large-scale community-level water defluoridation units typically rely on adsorptive processes such as the use of activated alumina sorbent and/or coagulation-precipitation processes such as the $\text{Al}(\text{OH})_3$ -based Nalgonda technique.^{7,8} Though Al-based F removal processes are effective in F removal, they generate toxic sludge containing high amounts of F and Al and leave residual Al in the treated water in concentrations high enough to pose risks of neurological disorders.^{9,10} As an alternative to Al-based F removal techniques, calcium-phosphate (Ca-PO_4)-based techniques^{11–16} are promising due to the low solubility of Ca-PO_4 in the prevalent groundwater pH range ($6\text{--}8$).^{13,16} Also, the end product, fluorapatite (FA; $\text{Ca}_5(\text{PO}_4)_3\text{F}$), is the most stable among the Ca-PO_4 solid phases and is nontoxic,¹⁷ which is why this process does not result in hazardous sludge formation. Fluorapatite is also relevant in biomedical studies as

formation of fluorapatite helps in preventing dental caries and tooth decay.¹⁸ Fluorapatite is also used to treat bone defects.¹⁹

Formation of FA using various Ca-PO_4 -based amendments has been extensively studied.^{14,15,20–26} Solids such as brushite ($\text{CaHPO}_4\cdot 2\text{H}_2\text{O}_{(s)}$),^{21,27} hydroxyapatite (HA; $\text{Ca}_5(\text{PO}_4)_3\text{OH}_{(s)}$),^{14,15,20,25,28–32} octa-calcium phosphate ($\text{Ca}_8\text{H}_2(\text{PO}_4)_6\cdot 5\text{H}_2\text{O}_{(s)}$),³³ and tricalcium phosphate ($\text{Ca}_3(\text{PO}_4)_2_{(s)}$)²⁴ were used to remove F from water. However, these methods have attributed the uptake of F as adsorption onto the solid surface as opposed to the precipitation of fluorapatite. Some studies have demonstrated the precipitation of FA using Ca, PO_4 , and F sources under conditions of initial

Received: April 30, 2024

Revised: September 13, 2024

Accepted: September 16, 2024